

## Chunking pitfalls, worked on one document

the same 1 100-word guest information sheet, split five different ways – and the question each split cannot answer

**The source document** – da-mario-guest-info.md, the one file we ingest. It has four sections, one of them a table:

### da-mario-guest-info.md (abridged)

```
# Da Mario - guest information

## Opening hours
Tue-Thu 17:00-23:00, Fri-Sat 17:00-24:00,
Sun 12:00-22:00. Monday closed.

## Delivery
We deliver within Saint-Étienne. Last order one
hour before closing. Delivery until 22:30 on Sundays.

## House dressing
Our house dressing is made fresh every day.
Ingredients: olive oil, lemon juice, Dijon mustard,
honey, salt, pepper. It contains no milk. It is not
vegan, because of the honey.

## Allergens per pizza
| pizza | gluten | milk | nuts |
| Margherita | yes | yes | no |
| Marinara | yes | no | no |
| Verdure | yes | no | no |
...19 more rows ...
```

**The splitter** – fixed windows, the ILaaS defaults from the slide:

### ingest settings

```
chunk_size = 800 # characters
chunk_overlap = 100 # characters
splitter = "fixed window"
```

**How to use these pages.** Each of the next five pages is one pitfall: the chunks exactly as the splitter produced them, the question a guest asks, the answer the process gives, and the one change that fixes it. Page 7 shows the same document chunked with its structure, and the same five questions answered.

**The rule behind all five:** a chunk is a **retrievable unit**. If it cannot be understood on its own, no retriever and no prompt can repair it later – and the citation it produces is not verifiable.

**Chunking is a data decision.** Document it with the collection: splitter, size, overlap, and the version of the document it was applied to. Two collections built with different settings are two different systems.

## Pitfall 1 – a table split across two chunks

the 800-character window falls inside the allergen table: one chunk has the header, the other has the rows

### chunk 4 (characters 2 400–3 200)

```
## Allergens per pizza
| pizza | gluten | milk | nuts |
| Margherita | yes | yes | no |
| Marinara | yes | no | no |
```

### chunk 5 (characters 3 100–3 900)

```
| Verdure | yes | no | no |
| Rucola | yes | yes | yes |
| Ortolana | yes | yes | no |
| Bufala | yes | yes | no |
```

**The guest asks:** “Does the Verdure contain milk?”

**What happens.** Retrieval returns chunk 5 – it is the one that mentions the Verdure. The passage says | Verdure | yes | no | no | and nothing else: the three values have no column names in this chunk. A model that reads the first one as the milk column answers “yes”, and the truth is “no”. Chunk 4, which carries the header, does not mention the Verdure at all, so it is never retrieved.

**Worse, quietly:** ask the same question about the *Rucola* – | Rucola | yes | yes | yes | – and every guess is right. The same broken chunk passes your test set and fails in production.

**The fix.** Split on row boundaries and repeat the header in every chunk that carries rows. A table row is the smallest self-contained statement of a table – with its header, never without.

## Pitfall 2 – a chunk without its heading

the window starts after “## House dressing”, so the chunk no longer says what it is about

### chunk 3 (characters 1 600–2 400)

Ingredients: olive oil, lemon juice, Dijon mustard, honey, salt, pepper. It contains no milk. It is not vegan, because of the honey.

**The guest asks:** “What is in the house dressing?”

**And the harder one:** “Is anything on your menu made with honey?”

**What happens.** The words “house dressing” are not in this chunk, so the embedding of the chunk is about *ingredients in general*. The question about the dressing may retrieve the wrong passage entirely – and if this chunk is retrieved, the citation says “document: guest-info, characters 1 600–2 400”, which no guest and no auditor can check.

**The second question** is answered correctly by accident: the honey is in the chunk, but the process cannot say *what* is made with honey, because the subject was cut away.

**The fix.** Prefix every chunk with its heading path and keep it in the metadata, not only in the text:

```
# Da Mario > House dressing
Ingredients: olive oil, ...
```

The heading is what makes the chunk retrievable *and* what makes the citation readable.

## Pitfall 3 – too small: the statement loses its subject

a 120-character window cuts between two sentences, and the second one starts with “It”

### chunk 12 (120 characters)

Our house dressing is made fresh every day.

### chunk 13 (120 characters)

It contains no milk.

**The guest asks:** “Is the Margherita free of milk?”

**What happens.** Chunk 13 is a short, confident, perfectly grounded sentence about *nothing*. It matches any milk question in the collection. The model answers “Yes – it contains no milk”, cites chunk 13, and the citation *supports* the sentence: the passage really does say that. The Margherita is made with mozzarella, so the answer is wrong – and the evidence looks impeccable.

**Why this is the dangerous one:** pitfalls 1 and 2 produce visible confusion. This one produces a fluent answer with a valid-looking citation.

**The fix.** A minimum chunk size, and never split inside a paragraph whose sentences refer to each other. If a chunk starts with “It”, “They”, “This” or “The same applies”, it is not a retrievable unit – merge it with the one before.

## Pitfall 4 – too large: the passage cites a page, not a fact

one chunk per document: everything is retrieved, nothing is quotable

### chunk 1 (4 800 characters – the whole sheet)

```
# Da Mario - guest information
## Opening hours
Tue-Thu 17:00-23:00, Fri-Sat 17:00-24:00,
Sun 12:00-22:00. Monday closed.
## Delivery
We deliver within Saint-Étienne. Last order
one hour before closing. Delivery until 22:30
on Sundays.
## House dressing ...
## Allergens per pizza ...
...
```

**The guest asks:** “Until when do you deliver on Sunday?”

**What happens.** The chunk is retrieved for every question about the restaurant, so the retrieval score says nothing: the top hit is the same passage whatever is asked, and *no hit* becomes impossible to detect. The model must find the answer inside 4 800 characters, and here two numbers compete – “Sun 12:00–22:00” and “delivery until 22:30 on Sundays”.

**And the explanation collapses:** the citation is “guest-info.md”, the whole document. A guest cannot verify it, an auditor cannot either, and your evidence test can only assert that *some* document was used.

**The fix.** Chunk down to one self-contained statement and keep its exact position (section, character offset) in the metadata, so the citation can quote the line it rests on – weak provenance, but pointing at a fact instead of a file.

## Pitfall 5 – the overlap duplicates a statement, and the duplicates age apart

`chunk_overlap = 100` puts the same sentence in two chunks – then the hours change

### chunk 2 (ends at character 1 700)

...Last order one hour before closing.

Delivery until 22:30 on Sundays.

### chunk 3 (starts at character 1 600)

Delivery until 22:30 on Sundays.

## House dressing ...

**The guest asks:** “Do you deliver late on Sunday?”

**What happens now.** Top- $k$  returns both chunks. The answer cites *two passages* that are one sentence, and a reader – or a grounding check that counts supporting passages – reads that as corroboration. The measurement is inflated by the splitter, not by the data.

**What happens in six weeks.** The sheet is edited to “until 21:30” and only the changed section is re-ingested. One copy is updated, the other is not, and the collection now holds two contradicting passages of equal score. The process answers whichever ranks higher today.

**The fix.** Deduplicate by content hash at ingest, and re-ingest **whole documents**, never single chunks: delete the document’s chunks, write them again, record the document version in the collection.

## The same document, chunked with its structure

headings kept, table rows kept whole, one self-contained statement per chunk – and the five questions answered

### chunk A

```
# Da Mario > Delivery
We deliver within Saint-Étienne. Last order
one hour before closing. Delivery until 22:30
on Sundays.
[doc=guest-info v7, section=Delivery]
```

### chunk B

```
# Da Mario > House dressing
Our house dressing is made fresh every day.
Ingredients: olive oil, lemon juice, Dijon
mustard, honey, salt, pepper. It contains no
milk. It is not vegan, because of the honey.
[doc=guest-info v7, section=House dressing]
```

### chunk C (one row, header repeated)

```
# Da Mario > Allergens per pizza
| pizza | gluten | milk | nuts |
| Verdure | yes | no | no | [doc=guest-info v7]
```

### What the five questions do now.

- 1 “Does the Verdure contain milk?” → chunk C: the row *and* its header – “no”, cited as one row.
- 2 “What is in the house dressing?” → chunk B, found by its heading, cited as a section.
- 3 “Is the Margherita free of milk?” → the Margherita row, not a sentence about the dressing.
- 4 “Until when do you deliver on Sunday?” → chunk A, one statement, quotable.
- 5 “Do you deliver late on Sunday?” → chunk A once, with the document version behind it.

**And what has not changed:** none of this makes the answer *true*. It makes the passage **quotable** and the failure **visible** – which is what the rest of lecture 6 measures.